Precise definition of a limit. Let f be defined on an open interval containing a but perhaps not at a itself. Then $\lim_{x\to a} f(x) = L$ means that given any $\varepsilon > 0$, we can find a corresponding $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } 0 < |x - a| < \delta.$$

1. Let c be a real number, and consider the constant function $f(x) = c$. We will prove that, if a is any real number, then $\lim_{x\to a} c = c$.

   (a) Using the animation at https://www.desmos.com/calculator/0e7jqsgfmr, use the precise limit definition to explain why $\lim_{x\to 2} 2 = 2$. That is, given any ε, explain how to find at least one $\delta > 0$ such that $f(x)$ is between the red horizontal dashed lines whenever x is between the blue vertical dashed lines.

   (b) Now write a proof that $\lim_{x\to a} c = c$ by filling in the blanks below:
   Let $\varepsilon >$ _____.
   Taking $f(x) = c$ and $L = c$, $|f(x) - L| =$ _____.
   Choose $\delta =$ _____. Then if $0 < |x - a| < \delta$, then _____ $< \varepsilon$. □

2. Now consider the function $f(x) = x$.

   (a) Using the animation at https://www.desmos.com/calculator/krhys6hi3a, use the precise limit definition to explain why $\lim_{x\to 5} x = 5$. For 3-4 values of ε, find the corresponding value of δ, recording your answers in a table. Do you see any pattern?

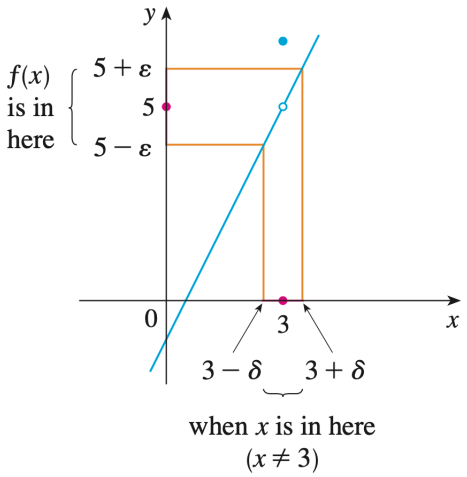
   (b) Now write a proof that $\lim_{x\to a} x = a$ by filling in the blanks below:
   Let $\varepsilon >$ _____.
   Taking $f(x) = x$ and $L = a$, $|f(x) - L| =$ _____.
   Choose $\delta =$ _____. Then _____. □

Example

Consider the function

$$f(x) = \begin{cases} 2x-1 & \text{if } x \neq 3, \\ 6 & \text{if } x = 3. \end{cases}$$

From the graph below, it looks like $\lim_{x\to 3} f(x) = 5$. To prove that $\lim_{x\to 3} f(x) = 5$, we need to show that for any $\varepsilon > 0$, we can find a corresponding $\delta > 0$ such that $|f(x) - 5| < \varepsilon$ whenever $0 < |x-3| < \delta$.



Step 1 - Preliminary Analysis

We begin by using the expression for $f(x)$ to write out $|f(x) - 5|$, which is the quantity we want to make small. For $x \neq 3$, $f(x) = 2x - 1$, so we have

$$\begin{aligned} |f(x) - 5| &= |2x - 1 - 5| \\ &= |2x - 6| \\ &= |2(x-3)| \\ &= |2||x-3| \\ &= 2|x-3|. \end{aligned}$$

Step 2 - Choose δ

The last equation above says that the distance between $f(x)$ and 5 is exactly twice the distance between x and 3. Thus, if we want to make $|f(x) - 5| < \varepsilon$, then we should take $|x-3| < \frac{\varepsilon}{2}$. We therefore choose $\delta = \frac{\varepsilon}{2}$.

Step 3 - Show that this choice of δ works

Suppose now that $0 < |x-3| < \delta$. Multiplying each side of the second inequality by 2 gives

$$2|x-3| < 2\delta.$$

But $2|x-3| = |f(x)-5|$ and $\delta = \frac{\varepsilon}{2}$, so this equation says precisely that

$$|f(x)-5| < 2\frac{\varepsilon}{2} = \varepsilon.$$

Step 4 - Conclude

Given $\varepsilon > 0$, we have found $\delta > 0$ such that $0 < |x-3| < \delta$ implies $|f(x)-5| < \varepsilon$. This proves that $\lim_{x\to 3} f(x) = 5$.

Exercises

1. Consider the function

$$f(x) = \begin{cases} 3x-2 & \text{if } x \neq 2, \\ 11 & \text{if } x = 3. \end{cases}$$

Follow the strategy above to prove that $\lim_{x\to 2} f(x) = 4$.

2. Prove that $\lim_{x\to a}(mx+b) = ma+b$ for any real numbers m, a, and b.

3. Prove that if $a > 0$, then $\lim_{x\to a}\sqrt{x} = \sqrt{a}$.